10TH CLASS MATHEMATICS — NEW SYLLABUS (PECTAA)

Exercise 12.2 — Probability (Addition Law)

Unit 12 | Exercise 12.2 | New Book (PECTAA)

Exercise 12.2 (Solved)

The whole exercise is the addition law: P(A ∪ B) = P(A) + P(B) − P(A ∩ B). When the two events cannot happen together, P(A ∩ B) = 0 and we only add.

1. If A and B are two events such that P(A) = 1/4, P(B) = 1/2 and P(A ∩ B) = 1/8, find P(A ∪ B).

P(A ∪ B) = P(A) + P(B) − P(A ∩ B)

P(A ∪ B) = 1 / 4 + 1 / 2 − 1 / 8

P(A ∪ B) = 2 / 8 + 4 / 8 − 1 / 8 = 5 / 8

Answer: P(A ∪ B) = 5/8

2. In an apartment, selecting a house from door numbers 1 to 50 randomly, find the probability of getting the door number of the house to be an even number or a perfect square number.

S = {1, 2, 3, ..., 50}, n(S) = 50

Even numbers A = {2, 4, 6, ..., 50}, n(A) = 25

Perfect squares B = {1, 4, 9, 16, 25, 36, 49}, n(B) = 7

A ∩ B = {4, 16, 36}, n(A ∩ B) = 3

P(A) = 25 / 50 = 1 / 2, P(B) = 7 / 50, P(A ∩ B) = 3 / 50

P(A ∪ B) = P(A) + P(B) − P(A ∩ B)

P(A ∪ B) = 25 / 50 + 7 / 50 − 3 / 50 = 29 / 50

Answer: P(even or perfect square) = 29/50

3. The probability of a team winning any match is 3/10 and the probability of losing any match is 2/10. What is the probability that (i) the team wins or loses a particular match (ii) the team neither wins nor loses a match.

Let W = winning a match, L = losing a match.

P(W) = 3 / 10, P(L) = 2 / 10

Winning and losing cannot happen together, so the two events are mutually exclusive.

(i) P(W ∪ L) = P(W) + P(L) = 3 / 10 + 2 / 10 = 5 / 10 = 1 / 2

(ii) P(neither wins nor loses) = 1 − P(W ∪ L) = 1 − 1 / 2 = 1 / 2

Answer: (i) P(win or lose) = 1/2 (ii) P(neither) = 1/2

4. In a single throw of two dice, find the probability of having sum of 7 or 11.

n(S) = 36

A = sum 7 = {(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)}, n(A) = 6

B = sum 11 = {(5, 6), (6, 5)}, n(B) = 2

A ∩ B = φ, so the events are mutually exclusive.

$P(A \cup B) = P(A) + P(B) = 6 / 36 + 2 / 36 = 8 / 36 = 2 / 9$

Answer: P(sum 7 or 11) = 2/9

5. Find the probability of getting a sum of 5 or 7 in a throw of two dice.

$n(S) = 36$

A = sum 5 = {(1, 4), (2, 3), (3, 2), (4, 1)}, $n(A) = 4$

B = sum 7 = {(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)}, $n(B) = 6$

The two events are mutually exclusive.

$P(A \cup B) = P(A) + P(B) = 4 / 36 + 6 / 36 = 10 / 36 = 5 / 18$

Answer: P(sum 5 or 7) = 5/18

6. A card is taken out at random from a standard pack of 52 cards. Find the probability of taking out (i) A king or a Jack (ii) Neither a king nor a Jack.

$n(S) = 52$

(i) Number of kings = 4, number of jacks = 4

$P(K \cup J) = P(K) + P(J) = 4 / 52 + 4 / 52 = 8 / 52 = 2 / 13$

(ii) $P(\text{neither}) = 1 - P(K \cup J) = 1 - 2 / 13 = 11 / 13$

Answer: (i) P(king or Jack) = 2/13 (ii) P(neither) = 11/13

7. A dice is thrown twice. What is the probability that at least one of the two throws comes up with number 3.

$n(S) = 36$

A = number 3 on the first throw = 6 outcomes

B = number 3 on the second throw = 6 outcomes

$A \cap B = \{(3, 3)\}$, so $n(A \cap B) = 1$

$P(A \cup B) = 6 / 36 + 6 / 36 - 1 / 36 = 11 / 36$

Or directly: $P(\text{at least one 3}) = 1 - P(\text{no 3 on both}) = 1 - (5 / 6)(5 / 6) = 1 - 25 / 36 = 11 / 36$

Answer: P(at least one 3) = 11/36

8. There are 15 cards in a bag marked as 1, 2, 3, …, 15. Find the probability of picking a card at random, the number written on which is a multiple of 5 or of 7.

$S = \{1, 2, 3, \ldots, 15\}$, $n(S) = 15$

Multiples of 5: {5, 10, 15}, $n(A) = 3$

Multiples of 7: {7, 14}, $n(B) = 2$

There is no common card, so $A \cap B = \varphi$.

$P(A \cup B) = 3 / 15 + 2 / 15 = 5 / 15 = 1 / 3$

Answer: P(multiple of 5 or 7) = 1/3

9. Two fair coins are tossed once. What is the probability of getting at least one head or two heads.

$S = \{HH, HT, TH, TT\}$, $n(S) = 4$

At least one head $A = \{HH, HT, TH\}$, $n(A) = 3$

Two heads $B = \{HH\}$, $n(B) = 1$

$A \cap B = \{HH\}$, $n(A \cap B) = 1$

$P(A \cup B) = 3/4 + 1/4 - 1/4 = 3/4$

The event “two heads” is already inside “at least one head”.

Answer: P(at least one head or two heads) = 3/4

10. At a busy intersection, 50% of vehicles turn right, 30% turn left and 20% go straight. What is the probability that a randomly selected vehicle turns left or right?

$P(\text{right}) = 50/100 = 0.5$, $P(\text{left}) = 30/100 = 0.3$

A vehicle cannot turn left and right at the same time, so the events are mutually exclusive.

$P(\text{left or right}) = 0.3 + 0.5 = 0.8$

Answer: P(left or right) = 0.8 = 80%

11. Two fair coins are tossed. What is the probability of getting either two heads or two tails?

$S = \{HH, HT, TH, TT\}$, $n(S) = 4$

Two heads $A = \{HH\}$, two tails $B = \{TT\}$

$A \cap B = \varphi$, so the events are mutually exclusive.

$P(A \cup B) = 1/4 + 1/4 = 2/4 = 1/2$

Answer: P(two heads or two tails) = 1/2

12. If A and B are two mutually exclusive events of a random experiment and P(not A) = 0.45, P(A ∪ B) = 0.65, then find P(B).

$P(A) = 1 - P(\text{not } A) = 1 - 0.45 = 0.55$

A and B are mutually exclusive, so $P(A \cup B) = P(A) + P(B)$

$0.65 = 0.55 + P(B)$

$P(B) = 0.65 - 0.55 = 0.10$

Answer: P(B) = 0.10

13. If P(A) = 2/3, P(B) = 2/5, P(A ∪ B) = 1/3 then find P(A ∩ B).

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$1/3 = 2/3 + 2/5 - P(A \cap B)$

$P(A \cap B) = 2/3 + 2/5 - 1/3$

$P(A \cap B) = 10/15 + 6/15 - 5/15 = 11/15$

Answer: P(A ∩ B) = 11/15

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14. A and B are two events such that P(A) = 0.42, P(B) = 0.48, and P(A ∩ B) = 0.16. Find: (i) P(Ā) (ii) P(B) (iii) P(A ∪ B)

(i) $P(\bar{A}) = 1 - P(A) = 1 - 0.42 = 0.58$

(ii) $P(B) = 1 - P(B) = 1 - 0.48 = 0.52$

(iii) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$P(A \cup B) = 0.42 + 0.48 - 0.16 = 0.74$

Answer: (i) 0.58 (ii) 0.52 (iii) 0.74

Remember

Add both probabilities, then subtract the common part once.

When the events cannot happen together, $P(A \cap B) = 0$, so just add.

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